

The Variable Target Handicap System

Making every rack matter, for everyone at the table

14.1 STRAIGHT POOL · SEASON PROPOSAL

THE PROBLEM

Straight pool is unforgiving. When a club-level beginner sits down against someone who regularly runs 30+, the outcome is decided before the break. The beginner stops trying around ball 15. The strong player feels no pressure until the very end. Nobody has fun, and the gap never closes.

Traditional flat handicaps (just a ball spot) help at the edges but ignore how much a player has improved since they were seeded. They also treat all wins the same — a blowout and a nail-biter update nothing.

CORE IDEA

Instead of giving weak players extra balls, we give everyone a *different finish line* — one that adjusts automatically after every match based on real results.

HOW IT WORKS

1 — Every player has a rating (300–1500)

Players are seeded at registration based on an honest self-assessment. The rating then moves up or down after every match based on who won, who was expected to win, and by how much.

2 — Your rating sets your target for each match

Forget a fixed 100-ball game. Each player races to their own target number, calculated directly from their rating:

```
target = round( (20 + (rating - 300) ÷ 15) ÷ 5 ) × 5 → result is always a multiple  
of 5, between 20 and 100
```

The table below shows what that looks like at each skill level:

SKILL LEVEL	DESCRIPTION	RATING	RACE-TO TARGET
Beginner	Rarely runs 5+	525	35
Below Average	Occasional short runs	675	45
Average	Runs of 10–15	900	60
Above Average	Runs of 15–25	1125	75
Advanced	Regular 20+ runs	1350	90
Expert	Consistent 30+ runs	1500	100

A match is simply: whoever hits their target first wins. A beginner racing to 35 while the expert races to 100 — *in the same match* — is a genuinely competitive contest.

EXAMPLE MATCH

MATCH PREVIEW

Carlos

BELOW AVERAGE · RATING 680

45

RACES TO

VS

Diane

ADVANCED · RATING 1340

90

RACES TO

Carlos wins — reaches 45 first.

Diane finishes at 38 / 90 (42% completion)

Carlos pulled off an upset. Because Diane only reached 42% of her target, the system reads this as a blowout in Carlos's favor. Ratings swing hard, Carlos climbs, and his target rises to reflect his real ability. Next time they meet, Carlos races to 50. The gap is closing.

THE RATING ENGINE (ELO + MARGIN)

After each match, a three-step calculation updates both ratings:

```
Step 1 – How likely was the winner to win? (Elo expected score)  $E = 1 / (1 + 10^{((Loser\_rating - Winner\_rating) / 200)})$ 
Step 2 – How decisive was it? (Margin multiplier, ranges 1x to 2x)  $M = 1 + (1 - loser\_score / loser\_target)$ 
Step 3 – Rating swing for this match  $\Delta = round(32 \times M \times (1 - E))$ 
Winner gains  $\Delta$  · Loser drops  $\Delta$  · Ratings stay within 300–1500
```

- 1 **Upsets reward more.** When a lower-rated player wins, E is small, so $(1 - E)$ is large, and the swing is big. The system quickly corrects a bad seed.
- 2 **Blowouts swing harder.** If the loser only reached 30% of their target, $M \approx 1.7\times$. A close match (loser at 90%) moves M to just $1.1\times$. Dominant wins drive handicaps apart fast; close ones make small adjustments.
- 3 **Expected wins barely move the needle.** When the better player wins as expected, E is near 1.0, so $(1 - E)$ is tiny. A K -factor of 32 keeps the math stable over a full season.

WHY THIS SYSTEM?

- ✓ **No manual tinkering.** The league operator never has to argue about whether someone's handicap is fair. The match results speak for themselves.
- ✓ **Every ball counts for the weaker player.** When you're racing to 45 and you're at 40, you feel the pressure. Players who once conceded early now fight for every inning.
- ✓ **Self-correcting seeds.** Over-seeded a player? No problem — a few matches and the rating adjusts. The floor of 20 means even a losing streak can't leave someone without a competitive target.
- ✓ **Transparent to players.** Everyone can see their rating, their current target, and exactly what will happen to both numbers after the match — before the last ball drops.
- ✓ **Familiar format, new stakes.** It's still straight pool. Same rules, same fouls, same beauty. We're just changing what "winning the game" means for each player at the table.

QUICK REFERENCE

FOR THE LEAGUE OPERATOR

When adding a player: pick the seed tier that honestly describes where they are *today*, not their peak. A conservative seed means their first few wins will push the rating up fast — which is more fun than getting hammered early and quitting.

RECORDING A MATCH

You need three things: who played, who won, and what score the loser finished on. That's it. The system handles everything else.

TARGET FLOOR

No player's target can drop below 20. This protects struggling players from facing an impossibly small number while still giving the math room to work.